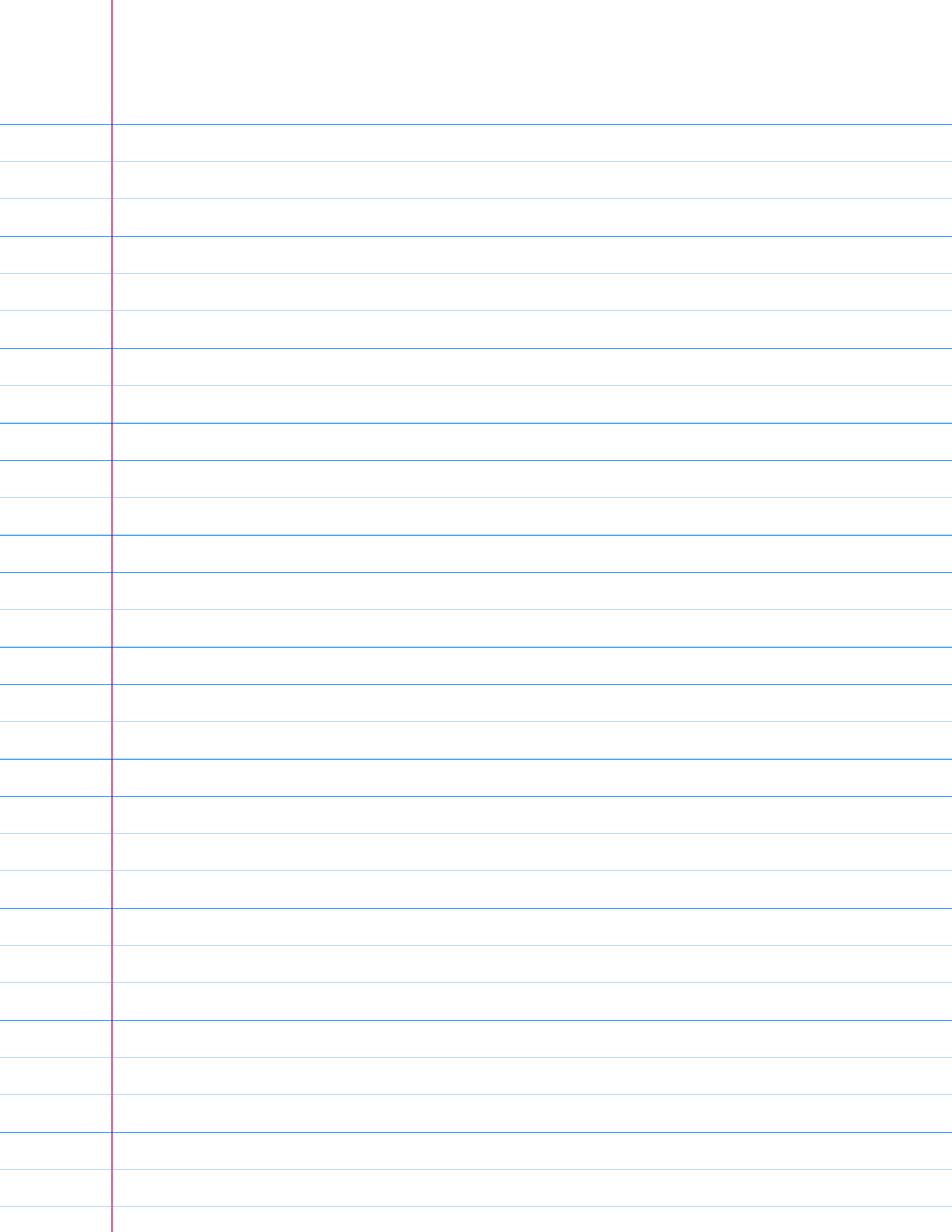




$$A\underline{x} = \underline{b}$$

$A: n \times p$ Every col of A is $n \times 1$.

$\underline{x}: p \times 1$ There are p cols in A .

$$\underline{b}: n \times 1$$

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} x_1 - x_2 + 0x_3 + 0x_4 &= 1 \\ 0x_1 + 1x_2 + 1x_3 + 1x_4 &= 0 \\ -x_1 + 0x_2 + 1x_3 - 1x_4 &= 0 \end{aligned}$$

Rm elimination on $\begin{pmatrix} 1 & -1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ -1 & 0 & 1 & -1 & 0 \end{pmatrix}$

If last col is free: then $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ is a linear comb of prior cols, i.e. cols of A . There is a soln.

If last col is pivot: then $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ is not a linear comb of prior cols, i.e. No SOLUTION.

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} c_4 &= c_1 + c_2 \\ -c_1 - c_2 + c_4 &= 0 \\ -1c_1 - 1c_2 + 0c_3 + 1c_4 &= 0 \end{aligned}$$

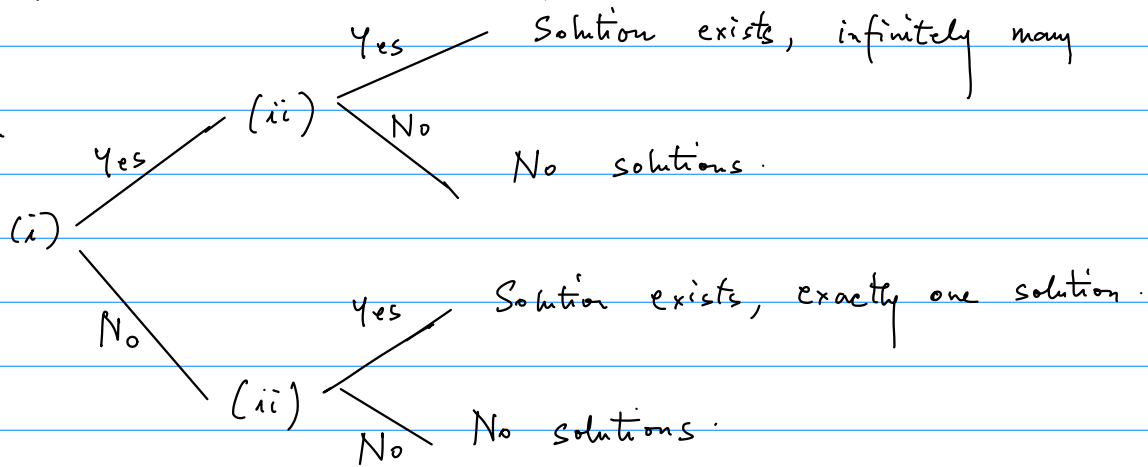
$$A \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad A\underline{x}^* = \underline{b}$$

$$A \left(\underline{x}^* + \beta \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right) = A\underline{x}^* + A\beta \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} = A\underline{x}^* + \beta \left(A \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right) = \underline{b} + \beta \underline{0} = \underline{b}$$

$$A\mathbf{x} = \mathbf{b}.$$

(i) Does A have free columns?

(ii) Is the last col a free col in $[A \ \mathbf{b}]$.



Suppose we found a pivot in every row of A before we exhaust all cols of A , then in $[A \ \mathbf{b}]$, the last col has to be free.

If A is square and we found a pivot in each row, then A has no free cols and $A\mathbf{x} = \mathbf{b}$ has a UNIQUE soln for each $\mathbf{b}$.

* If A is wide $\left[\begin{matrix} \# \text{rows} < \# \text{cols} \end{matrix} \right]$, then if we find a pivot in each row, then $A\mathbf{x} = \mathbf{b}$ has infinitely many solns no matter what $\mathbf{b}$ is.

* If A is tall $\left[\begin{matrix} \# \text{rows} > \\ \# \text{cols} \end{matrix} \right]$, and if each col is a pivot col, then $A\mathbf{x} = \mathbf{b}$ has $\begin{cases} \text{unique soln if } \mathbf{b} \text{ is a linear comb of cols of } A. \\ \text{No soln if } \mathbf{b} \text{ is NOT a linear comb of cols of } A. \end{cases}$